



Project Presentation: Pokémon GO Raid Bot

This presentation provides a technical overview of a Telegram bot developed for the Pokémon GO community. We will explore its architecture, functionalities, and future integrations.

The Project's Core: Node.js Backend and Telegram

The project is based on a robust Node.js backend, designed to manage interactions with the Telegram platform.

Communication takes place via the Telegram API, allowing the bot to receive commands from users and respond in real-time.

It uses the **Grammy** library, an excellent choice for its flexibility and powerful features, which simplify the development of complex and scalable bots.

```
export function raidCommand(bot) {
  bot.command("raid", async (ctx) => {
    // fetchBosses();

    const args = ctx.match?.split(" ") || [];

    if (args.length < 3) {
      //           0           1           2           3           4
      return ctx.reply("Uso: /raid [pokemon] [gym] [ora inizio] [ora fine?] [note?];");
    }

    const pokemon = args[0] || "";
    const gym = findGymByKeywords(args[1]);
    const timePattern = /^[0-1]?[0-9]|2[0-3]:[0-5][0-9]$/;
    const start = args[2] && timePattern.test(args[2]) ? args[2] : new Date().toString();
    let end = "";
    let notes = "";
    if (args[3] && timePattern.test(args[3])) {
      end = args[3];
      notes = args.slice(4).join(" ");
    } else {
      notes = args.slice(3).join(" ");
    }

    if (!gym) return ctx.reply("Nome palestra errato, riprova con un nuovo comando.");

    const raid = new Raid({
      pokemon: pokemon,
      gym:      gym.nome,
      lat:      gym.lat,
      lon:      gym.lon,
      start:    start,
      end:      end,
      notes:    notes,
      creator:  ctx.from?.first_name
    });
  });
}
```

Interactive Commands: Simplifying Raid Management

The bot is designed to simplify the organization and management of Pokémon GO Raids through intuitive commands directly from Telegram. Each command is structured to efficiently collect essential information.

1

**/raid [pokemon] [location]
[start time] [end time]
[notes]**

Allows users to announce a new raid, specifying all necessary details for other players.

2

/info [pokemon]

Provides quick and detailed information about a specific Pokémon, useful for preparing for raids or for quick reference.

3

Usage Examples

Users can easily create and view events, improving community coordination.



Architecture and Technology Stack

The heart of the system is a Node.js application, which leverages asynchronous programming to handle a large number of concurrent requests. The modularity of the code ensures ease of maintenance and scalability.

Node.js

Server-side JavaScript runtime environment for high-performance backend.

Grammy.js

Modern and efficient framework for Telegram bot development.

Telegram API

Direct interface for communication with the messaging platform.

Future Developments: The React.js User Interface

In addition to the Telegram bot, a Single Page Application (SPA) in React.js is under development. This web interface aims to offer a more advanced and interactive visualization of the data generated by the bot.

Planned Features:

- **Command History:** Detailed view of all executed commands.
- **Bot Usage Ranking:** Statistics on user interaction with the bot.
- **Data Analysis:** Reports and graphs on the most popular raids and Pokémon.
- **Advanced Management:** Tools for maintainers to monitor bot activity.





Contributions and Future Directions

The project is open-source and welcomes contributions from the community. The public repository serves as a starting point for anyone wishing to explore the code or propose improvements.



Public Repository

[Access the Code](#)



Collaboration

We are open to pull requests and new ideas.



Evolution

The bot is continuously evolving to support new features and enhance user experience.